

Physical Layer

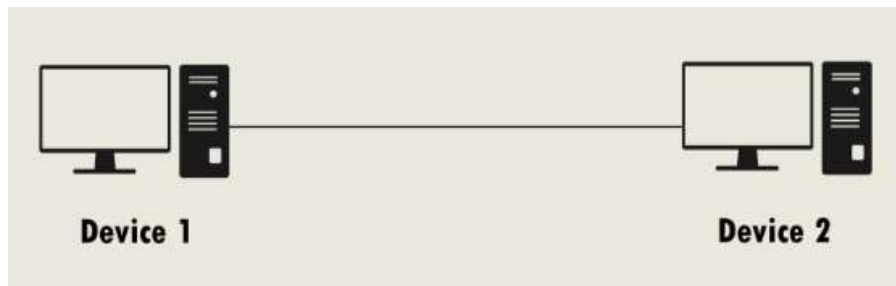
- This section focuses on the physical aspects of data transmission, which is the very foundation of network communication.
- We will look into different network layouts (topologies), how data signals travel across various media and the modes used for transmission.

Types of Network Topology

Network Topology = how devices (nodes) are connected in a network.

1) Point-to-Point Topology

Two devices connected directly, simplest link.

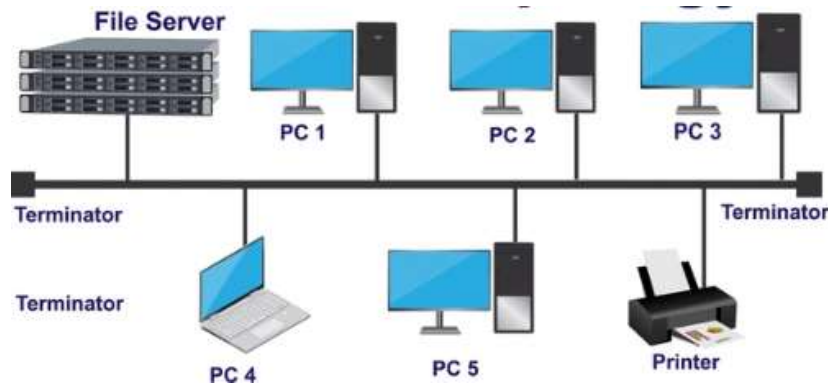


Points

- Only two devices connected.
- Fast and simple.
- Not scalable (only 2 devices).

2) Bus Topology

All devices share one main cable (backbone).

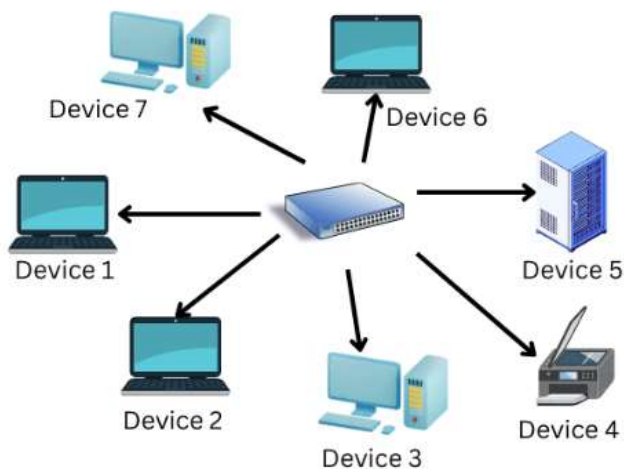


Points

- One main cable connects all.
- Cheap and easy to set up.
- If the main cable fails, the whole network fails.

3) Star Topology

Each device connects to a central hub.

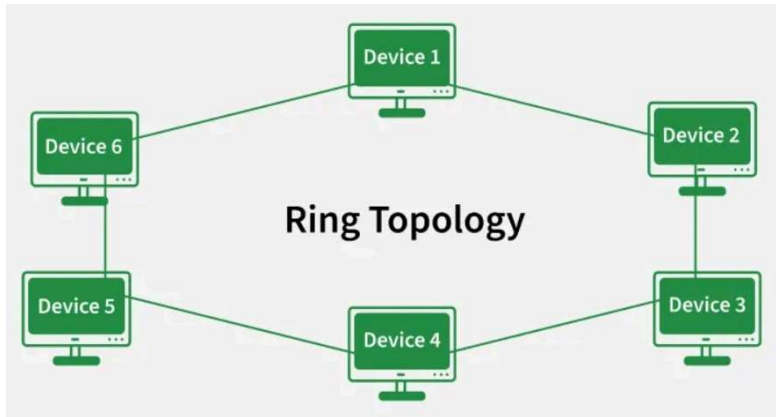


Points

- Every device connects to the hub/switch.
- Easy to add or remove devices.
- If the central hub fails, the whole network fails.

4) Ring Topology

Devices form a closed loop.

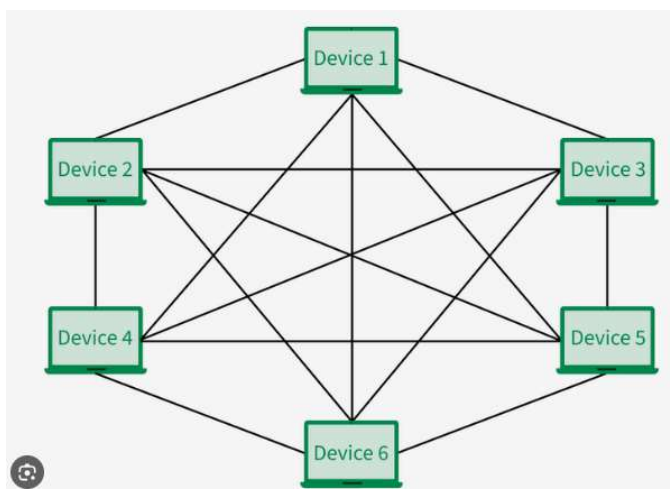


Points

- Data goes around in a circle.
- Less collision of data.
- One device failure can stop the network.

5) Mesh Topology

Every device connects directly to every other device.

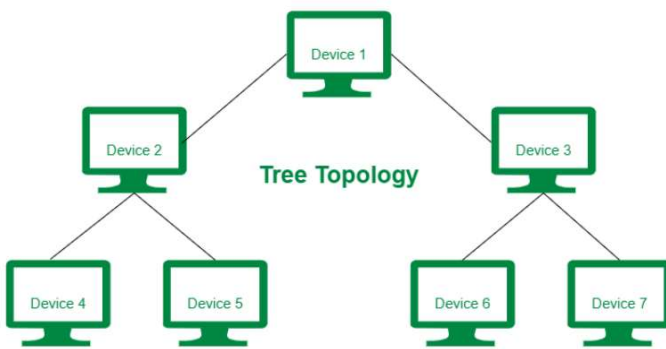


Points

- Full connection between devices.
- Very reliable — many paths.
- Lots of cables = expensive.

6) Tree Topology

Hierarchical (like a real tree).

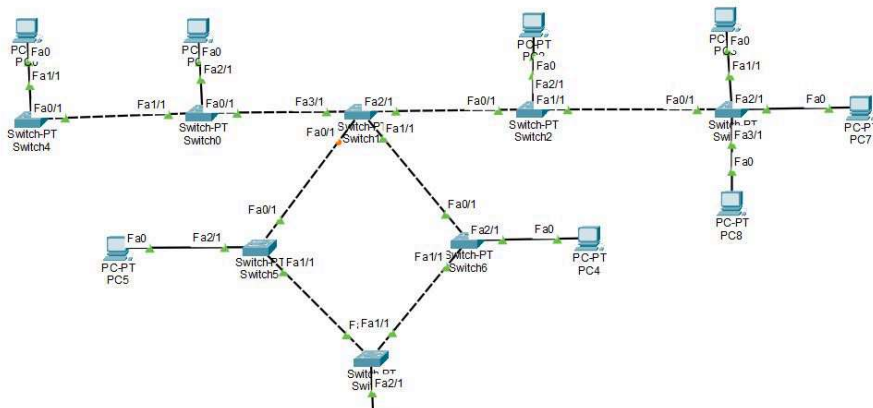
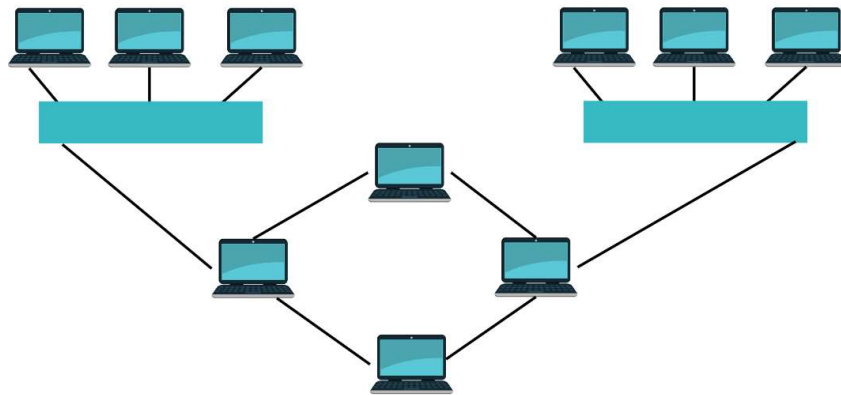


Points

- Central root + branches.
- Easy to expand.
- Backbone failure affects the network.

7) Hybrid Topology

Combination of two or more topologies.



Points

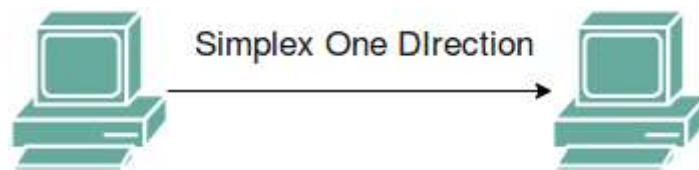
- Flexible and scalable.
- Can mix star, bus, ring, etc.
- Expensive and complex.

Transmission Modes

Transmission modes also known as communication modes, define how data is transferred between devices in a network or bus system. They are classified into three types: **Simplex Mode**, **Half-Duplex Mode**, and **Full-Duplex Mode**.

Simplex Mode

In Simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit, the other can only receive.



***Example:** Keyboard and traditional monitors. The keyboard can only introduce input, the monitor can only give the output.*

Pros of Simplex Mode

- It is the most cost-effective mode, as it only requires one communication channel.
- There is no need for coordination between the transmitting and receiving devices, which simplifies the communication process.
- Simplex mode is particularly useful in situations where feedback or response is not required, such as broadcasting or surveillance.

Cons of Simplex Mode

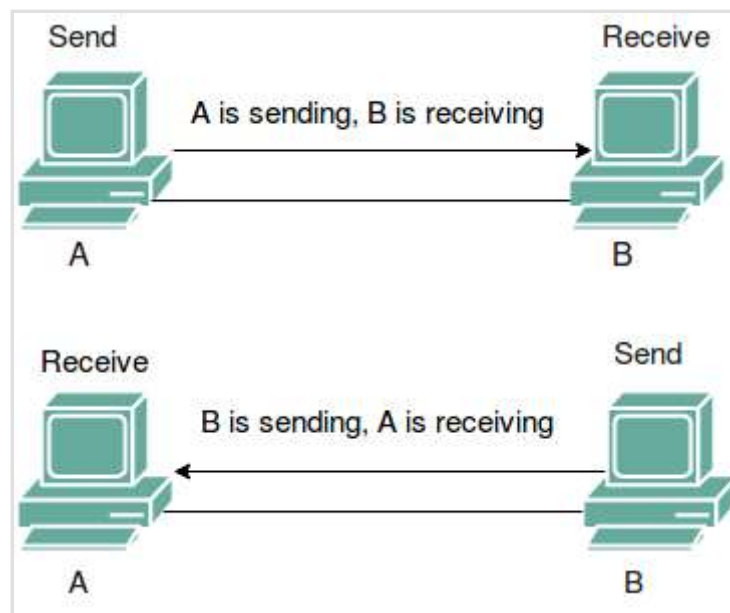
- Only one-way communication is possible.

- There is no way to verify if the transmitted data has been received correctly.
- Simplex mode is not suitable for applications that require bidirectional communication.

Half-Duplex Mode

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The half-duplex mode is used in cases where there is no need for communication in both directions at the same time. The entire capacity of the channel can be utilized for each direction.

$\text{Channel Capacity} = \text{Bandwidth} * \text{Propagation delay}$



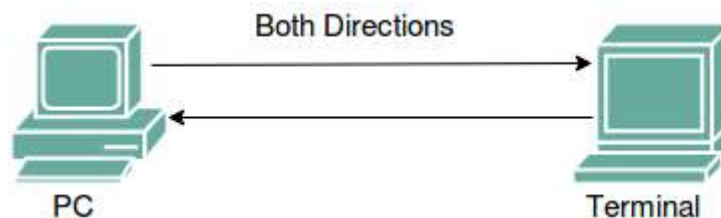
Example: Walkie-talkie in which messages are sent one at a time and messages are sent in both directions.

Full-Duplex Mode

In full-duplex mode, both stations can transmit and receive simultaneously. It is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions. Here, signals going in one direction share the capacity of the link with signals going in another direction, this sharing can occur in two ways:

- Either the link must contain two physically separate transmission paths, one for sending and the other for receiving.
- Or the capacity is divided between signals traveling in both directions.

Channel Capacity = $2 * \text{Bandwidth} * \text{propagation Delay}$



Example: Telephone Network in which there is communication between two persons by a telephone line, through which both can talk and listen at the same time.

Pros of Full-Duplex Mode

- Ideal for real-time applications such as video conferencing or online gaming.
- Provides a high level of reliability and accuracy, as there is no need for error correction mechanisms.

Cons of Full-Duplex Mode

- It's the most expensive mode, as it requires two communication channels.
- It is more complex than simplex and half-duplex modes, as it requires two physically separate transmission paths or a division of channel capacity.
- Full-duplex mode may not be suitable for all applications, as it requires a high level of bandwidth and may not be necessary for some types of communication.

Protocols in Physical Layer

Typically, a combination of hardware and software programming makes up the physical layer. It consists of several protocols that control data transmissions on a network. The following are some examples of Layer 1 protocols:

- **Ethernet (IEEE 802.3)** : Widely used for wired networks.
- **Wi-Fi (IEEE 802.11)** : For wireless communication.
- **Bluetooth (IEEE 802.15.1)** : Short-range wireless communication.
- **USB (Universal Serial Bus)** : For connecting devices over short distances.

What is IEEE?

IEEE stands for **Institute of Electrical and Electronics Engineers**.

- It's an international organization
- It **creates standards** so devices from different companies can work together
- When you see something like **IEEE 802.x**, it means:
"This technology follows a standard defined by IEEE"

What is IEEE 802?

IEEE 802 is a family of networking standards related to **Local Area Networks (LANs)** and **Personal Area Networks (PANs)**.

Each number after **802** defines a **specific type of network technology**.

1. Ethernet (IEEE 802.3)

Ethernet is the most common **wired network technology**.

Key points:

- Uses **cables** (like Ethernet/LAN cables)
- Very **fast and reliable**
- Common in offices, schools, homes

IEEE 802.3 means:

- **802** → LAN standards
- **.3** → Ethernet standard

Example uses:

- Connecting a PC to a router
- Office networks
- Data centers

2. Wi-Fi (IEEE 802.11)

Wi-Fi is used for **wireless local area networking**.

Key points:

- Uses **radio waves**
- No physical cables
- Slightly less stable than Ethernet but very convenient

IEEE 802.11 means:

- **802** → LAN standards
- **.11** → Wireless LAN (Wi-Fi)

Example uses:

- Mobile phones
- Laptops
- Smart TVs

3. Bluetooth (IEEE 802.15.1)

Bluetooth is for **short-range wireless communication**.

Key points:

- Works over **very short distances**
- Low power consumption
- Used for personal devices

IEEE 802.15.1 means:

- **802** → Network standards
- **.15** → Personal Area Networks (PAN)
- **.1** → Bluetooth

Example uses:

- Wireless headphones
- Keyboard & mouse

- File transfer between nearby devices

4. USB (Universal Serial Bus)

USB is **not** an IEEE 802 standard.

Key points:

- Used to **connect devices**.
- Works over **short distances**.
- Supports both **data transfer and power**.

Examples:

- USB flash drives
- Keyboard and mouse
- Mobile phone charging

USB versions:

- USB 2.0
- USB 3.0 / 3.1 / 3.2
- USB-C (connector type)